

Treat all the following questions:

- Present the advantages and disadvantages of object-oriented database. (2 pts)
- The company staff may be classified into two categories: The managers and the employees. In addition, we distinguish two sub-categories of employees: the workers and the secretaries.
A person is characterized by a number, a family name; a name and a date of birth characterize a person. In addition, a manager is characterized by his experience (number of years); a worker is characterized by a work contract type;
A secretary is characterized by a work contract type and the number of typed characters by minute. The possible values of work contract type are: permanent, weekly and monthly.
A worker can has many careers. A number and a name characterize a career.
The workers and the secretaries work under the supervision of managers.
A manager may be a supervisor of other managers or a supervisor of employees.

a. Construct a class diagram for this system.
b. Use an ODL to transform this class diagram to object database (5 pts)
- Giving the following DATALOG rules :

(R1): PARENT(x,y) $\leftarrow$ FATHER(x,y,t), t>18 where (x : father-name, y : child-name, t :age)
(R2): PARENT(x,y) $\leftarrow$ MOTHER(x,y,t), t>18 where (x : mother-name, y : child-name, t :age)
(R3) : GRANDPARENT(x,z) $\leftarrow$ PARENT(x,y), PARENT(y,z)
(R4) : PREDECESSOR(x,y) $\leftarrow$ PARENT(x,y)
(R5) : PREDECESSOR(x,z) $\leftarrow$ PREDECESSOR(x,y), PARENT(y,z)

a. Draw the relational tree/graph that presents PREDECESSOR of child NASSER using the given rules.
b. Add new rules that find fathers who work in a same enterprise. (5 pts)
- After various changes, the database originally placed in a single hospital has been evolved and now manages several hospitals in the Alsace region. The database schema is now the following:

Service (Snum, SName, Shopital, SBuilding, SDirector)
Room (Snum, Rnum, RSupervisor, R_Nb_beds)
Employee(ENum, EName, EAddress, ETelephone)
Doctor (Dnum, DSpeciality)
Nurse (Nnum, Snum, NRotation, NSalary)
Patient (Pnum, Snum, Rnum, Pbed, PName, PAddress, PTelephone, PMutual, PSupport)
Action (Dnum, Pnum, Date, ADescription, ACoefficient)

In this scheme, the service director is a doctor designated by number. The room number is local to a service in the same hospital (There are several rooms number of 12). A room's supervisor is a nurse designated by number. For a room, you know its number of beds. Doctors and nurses are employees. Their identification as doctors or nurses is the same as employees.

In fact, this database is distributed over three computer sites corresponding to the values of the attribute "hospital" of the relationship Service:

Site name	Value of the attribute "hospital"
Strasbourg	Ambroise-Paré
Colmar	Colmar
Regional	All the other values

Propose and justify a good decomposition of the hospital base on these three sites. For this, use the horizontal and vertical decomposition as well as replication, whereby:

The sites of Strasbourg and Colmar manage only the correspondent hospitals.

The nurses are employed in a given service.

Doctors often work on several hospitals or clinics.

The management of hospital beds is local to each hospital.

We want to gather the management of the hospitalization fees at the regional center.

For each obtained fragment, we give its definition as a calculation of the relational algebra from the initial scheme (global).

(8 pts)